

Darshan Thakur

Mechanical Design | CFD | FEA | Thermal Management | Additive Manufacturing

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Professional Summary

Mechanical Engineer with an MSc (1:1 Distinction) from Trinity College Dublin and 3+ years of industry experience in CFD simulation, FEA, thermal management, and additive manufacturing. Experienced in critical infrastructure cooling, HVAC simulation, and structural analysis. Currently leading 3D printing technology implementation at Irish Manufacturing Research (Prodieco) and designing pharmaceutical-grade tablet feeding systems.

Education

Trinity College Dublin, MSc Mechanical Engineering Sept 2024 – Sept 2025

- **Grade:** 1:1 (Distinction)
- **Focus Areas:** Advanced Product Development, Computational Analysis, and Zero-Carbon Technologies

Mumbai University, BE Mechanical Engineering July 2018 – August 2021

- **Grade:** 8.92 / 10
- **Focus Areas:** Finite Element Analysis, Industrial Automation, Mechanical Vibrations, Machine Design.

Professional Experience

Graduate AM Engineer, Irish Manufacturing Research (Prodieco) –Dublin November 2025 – Present

- Leading the evaluation and implementation of new 3D printing technologies within the organization, including DLP resin printing and FDM, creating project timelines, defining KPIs, and managing workflows from technology assessment through to full deployment
- Researching and exploring metal additive manufacturing technologies for future applications, assessing material capabilities, process parameters, and feasibility for pharmaceutical and industrial components.
- Designing and developing pharmaceutical-grade tablet feeding systems including Vertical Feeders, Drum Feeders, and Simtap Feeders, ensuring suitability for GMP environments and downstream integration.
- Redesigning existing conventional components for additive manufacturing, optimizing geometry, support structures, and material selection to improve printability, functionality, and cost efficiency.
- Supporting the design and development of parts across the full AM workflow from concept through prototyping and functional testing to production handover.
- Exploring and evaluating AM materials including resins, filaments, and engineering polymers suitable for pharmaceutical-grade applications in collaboration with R&D and manufacturing teams.
- Integrating AM solutions into existing product lines, working closely with design, R&D, and manufacturing teams to ensure seamless transitions from conventional to additive processes.
- Maintaining technical documentation for AM processes, design iterations, material selections, and process parameters to support quality and compliance requirements.

Junior Design Engineer, Ansyscad Solutions –India January 2022 – August 2024

- Delivered CFD simulations across large-scale critical infrastructure projects including military shelter HVAC systems, data center thermal management from server rack level to rack-scale UPS systems, and electronic enclosure cooling using Ansys Fluent and Simcenter FLOEFD, optimizing airflow distribution, ducting layouts, vent geometries, and cooling strategies to eliminate hotspots and meet defense-grade thermal requirements.
- Performed FEA and CFD analysis for industrial chillers, large scale industrial ovens and thermal systems, applying conjugate heat transfer analysis, heat exchanger performance evaluation, and fan placement optimization to validate designs across defense, energy, and industrial sectors.
- Performed comprehensive FEA including linear, non-linear, fatigue, modal, and vibration studies to validate structural integrity of complex assemblies under dynamic and static loading.
- Applied Multi Body Dynamic simulation to analyze kinematics, mechanism behavior, and dynamic loads across missile transporters, launchers, and military tanks; used DEM to simulate particle flow and bulk material handling in industrial equipment.

- Drove full concept-to-product development from initial concept and 3D modelling through to manufacturing drawings and production handover, creating detailed CAD models, BOM and GD&T engineering drawings compliant with ISO, ASME, and MIL-STD standards using SolidWorks, NX and AutoCAD.
- Optimized designs through material selection, topology refinement, and ribbing strategies achieving weight and cost reductions without compromising performance; coordinated with manufacturing, testing, and procurement teams to resolve design issues and deliver assemblies on schedule.

Key projects:

- Defense & Aerospace: Military shelters, Missile transporters, Missile launchers, Rail Transporters
- Transportation & Mobility: Trailers, Tippers, Military tank, Cranes
- Consumer Products: Pallets, Industrial ovens, and Chillers

Formula Student Team Member, Team MH08 Formula Racing – India January 2019 – August 2020

- Supported chassis design and material selection using CAD tools for performance and safety optimization.
- Assisted in composite manufacturing and assembly, gaining hands-on fabrication experience.
- Supported sponsorship acquisition and managed social media outreach, boosting team visibility and funding.

Key Skills

CAD & Additive Manufacturing: SolidWorks, AutoCAD, NX, Creo, Autodesk Inventor, FDM & DLP Resin printing

Simulation/Analysis Software: Ansys Workbench, Simcenter FLOEFD, Hypermesh, Motion View, Fusion 360

Design Standards: ISO, ASME, ASTM, MIL STD

Soft Skills: Technical writing, Project management, KPI development, Cross-functional team leadership, Client-facing design reviews.

Programming Language: MATLAB (Statistical analysis), Python (Simulation post-processing and data visualization)

Certifications & Memberships

- Certified SolidWorks Professional (CSWP), Dassault Systems
- Becoming a HVAC Professional, Alison
- Fundamentals of Project Management, eCollege
- Member, American Society of Heating, Refrigerating and Air Conditioning Engineers (ASHRAE)
- Member, Engineers Ireland

Research & Projects

Benchmarking and Optimizing Enhanced Natural Convection Heat Sinks

August 2025

MSc Dissertation – Trinity College Dublin

- Developed a standardized benchmarking methodology using DOE and HEEDS enabling true optimum-to-optimum comparisons between displacement fin and straight-fin heat sink designs under natural convection.
- Demonstrated 9.53% (small fins) and 11.09% (large fins) total heat transfer improvement; SV1 displacement fins achieved 5–6% higher mass-specific efficiency (Q/m), providing a credible design tool for passive cooling trade-off decisions in data centers and power electronics.

Selected Portfolio Projects:

- CryoSync GPU Liquid Cooling System: maintained GPU below 36°C with 85% energy reduction at low thermal load.
- UAV Propeller Noise Reduction: 31.96% thrust increase alongside 21.2% noise reduction through blade geometry optimisation.
- PortaVax Vaccine Cold Chain Cooler: power-free passive cooler maintaining below 8°C for 25 hours with 840-vial capacity.
- Wind Resource Assessment: statistical MATLAB analysis of wind energy potential for turbine siting.

Explore more innovative research projects at: thakurd9.github.io/Portfolio/